Best critical region example

(Based on 8.6-4, page 404)

$X_1, \dots, X_{16} \sim N(\mu, 36)$

a) Show that a uniformly most powerful critical region for testing $H_0: \mu = 50$ vs $H_1: \mu > 50$ can be designed using statistic $\bar{x}$.

b) Find a general formula for uniformly best critical regions

c) What is the uniformly most powerful crit. region of size $\alpha = 0.05$?

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a) Neyman-Pearson lemma for testing $H_0: \mu = 50$ vs $\hat{H}_1: \mu = \mu_1$, $\mu_1 > 50$ requires to bound $\frac{L(50)}{L(\mu_1)} \leq k$ ← some number

maximal likelihood function

We computed in class 02/29 (see also p. 404) that

$$\circledast \quad \boxed{\frac{L(50)}{L(\mu_1)} \leq k \quad \text{if and only if} \quad \bar{x} \geq \frac{\mu_1 + 50}{2} - \frac{\ln k \cdot 72}{16(\mu_1 - 50)}}$$

Since the expression depends only on statistic $\bar{x}$ (not on $X_1 \dots X_n$ in any other way) $\Rightarrow$ $\bar{x}$ is enough to define best critical region.

b) $\tilde{C} = \{\bar{X} \geq c\}$ is uniformly best critical region (for any c)

Why?

Fix any $\mu_1 > 50$. By definition of "uniformly best," it is enough to show that $\{\bar{X} \geq c\}$ is best for $H_0: \mu = 50$ vs $H_1: \mu = \mu_1$ (for any $\mu_1 > 50$).

It is enough to show that conditions of Neyman - Pearson lemma are satisfied.

Indeed, choose k, such that $c = \frac{\mu_1 + 50}{2} - \frac{72 \ln k}{16(\mu_1 - 50)}$,

or, $k = \exp\left(\left(\frac{\mu_1 + 50}{2} - c\right) \cdot \frac{16}{72}(\mu_1 - 50)\right)$.

Then $\frac{L(0.5)}{L(\mu_1)} \leq k$ if and only if $\{\bar{X} \geq c\} =$

$= \{(X_1 \dots X_n) \in \tilde{C}\}$

(by ⊛)

and $k \geq 0$ ($\exp(x) \geq 0$ always)

$\Rightarrow \tilde{C}$ is the best critical region for testing H_0 vs H_1 (for any c)

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c) Using the result of b), it is enough to find c, such that $\{\bar{X} \geq c\}$ has correct significance level (since all such critical regions are "best")

$P(\bar{X} \geq c) = 0.05 \iff P\left(Z \geq \frac{c - 50}{6/4}\right) = 0.05$, $Z \sim N(0,1)$

$(c-50)/6/4 \approx 1.645$ $\boxed{c \approx 52.47}$